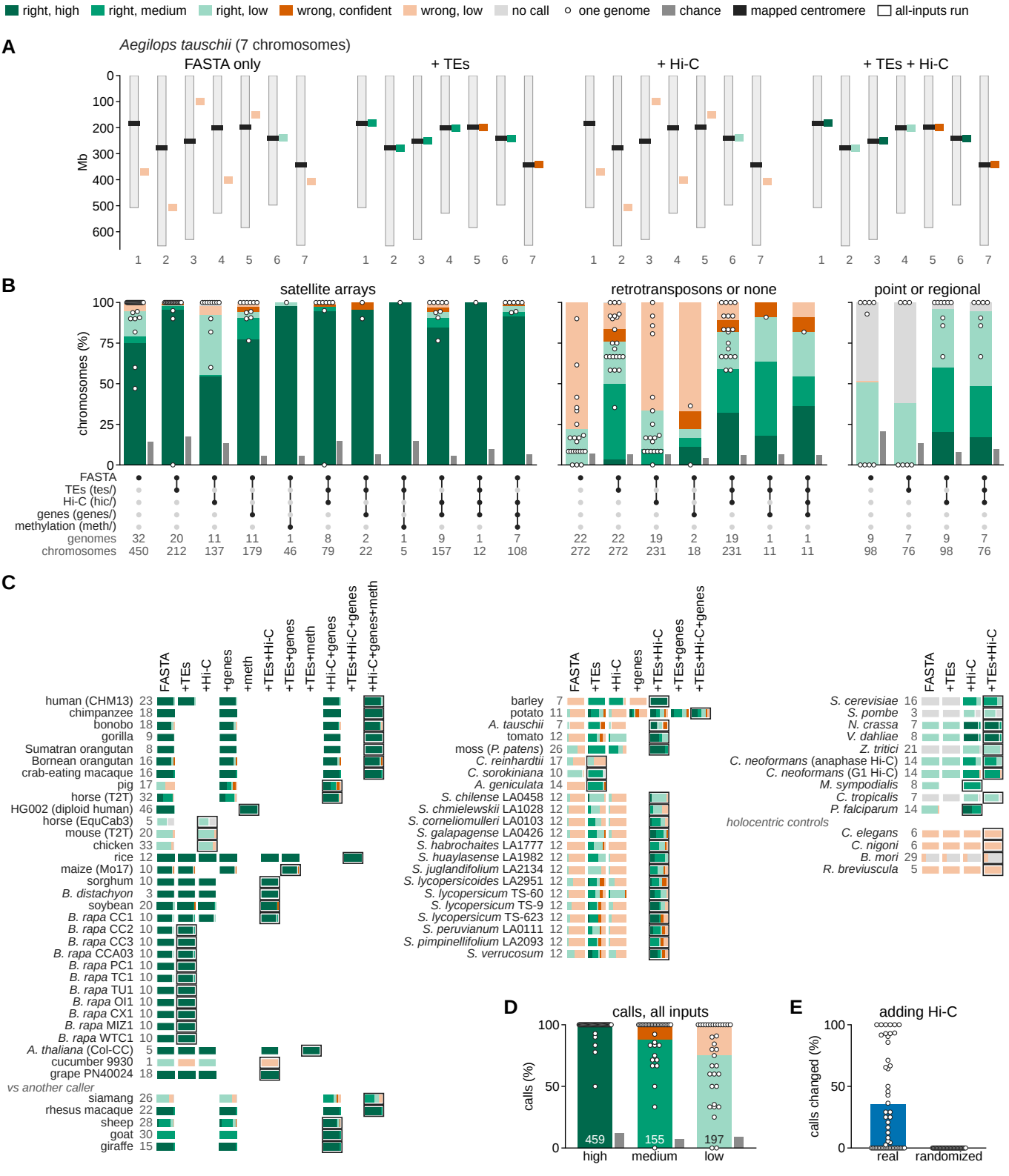

right, high
right, medium
right, low
wrong, confident
wrong, low
no call
one genome
chance
mapped centromere
all-inputs run
A
Aegilops tauschii (7 chromosomes)
FASTA only
+ TEs
+ Hi-C
+ TEs + Hi-C
Mb
B
satellite arrays
retrotransposons or none
point or regional
chromosomes (%)
FASTA
TEs (tes/)
Hi-C (hic/)
genes (genes/)
methylation (meth/)
genomes
chromosomes
C
FASTA
+TEs
+Hi-C
+genes
+meth
+TEs+Hi-C
+TEs+genes
+TEs+meth
+Hi-C+genes
+TEs+Hi-C+genes
+Hi-C+genes+meth
human (CHM13) 23
chimpanzee 18
bonobo 18
gorilla 9
Sumatran orangutan 8
Bornean orangutan 16
crab-eating macaque 16
pig 17
horse (T2T) 32
HG002 (diploid human) 46
horse (EquCab3) 5
mouse (T2T) 20
chicken 33
rice 12
maize (Mo17) 10
sorghum 10
B. distachyon 3
soybean 20
B. rapa CC1 10
B. rapa CC2 10
B. rapa CC3 10
B. rapa CCA03 10
B. rapa PC1 10
B. rapa TC1 10
B. rapa TU1 10
B. rapa OI1 10
B. rapa CX1 10
B. rapa MIZ1 10
B. rapa WTC1 10
A. thaliana (Col-CC) 5
cucumber 9930 1
grape PN40024 18
vs another caller
siamang 26
rhesus macaque 22
sheep 28
goat 30
giraffe 15
barley 7
potato 11
A. tauschii 7
tomato 12
moss (P. patens) 26
C. reinhardtii 17
C. sorokiniana 10
A. geniculata 14
S. chilense LA0458 12
S. chmielewskii LA1028 12
S. corneliomulleri LA0103 12
S. galapagense LA0426 12
S. habrochaites LA1777 12
S. huaylasense LA1982 12
S. juglandifolium LA2134 12
S. lycopersicoides LA2951 12
S. lycopersicum TS-60 12
S. lycopersicum TS-9 12
S. lycopersicum TS-623 12
S. peruvianum LA0111 12
S. pimpinellifolium LA2093 12
S. verrucosum 12
S. cerevisiae 16
S. pombe 3
N. crassa 7
V. dahliae 8
Z. tritici 21
C. neoformans (anaphase Hi-C) 14
C. neoformans (G1 Hi-C) 14
M. sympodialis 8
C. tropicalis 7
P. falciparum 14
holocentric controls
C. elegans 6
C. nigoni 6
B. mori 29
R. breviuscula 5
D
calls, all inputs
calls (%)
459
155
197
high
medium
low
E
adding Hi-C
calls changed (%)
real
randomized

What a centromere is made of decides which data find it. From the FASTA alone cenfuse was right on 94% of satellite centromeres. Centromeres of retrotransposons or of no repeat went from 22% to 76% with the TE annotation, and the point and regional centromeres of fungi and *Plasmodium* from 51% to 96% with Hi-C.

A One genome, *Aegilops tauschii*: its chromosomes to scale, its centromeres mapped by CENH3 (black) and the call made from each set of inputs, colored by outcome (small ones drawn as a visible mark). Its gain from the FASTA alone to all its inputs is nearest the median of the 34 genomes run with all four sets. A call is right within 1 Mb of the mapped centromere, so one beside it can still miss.

B Every chromosome with a mapped centromere, pooled by inputs and centromere type: right (green, darker when more confident), wrong (vermillion, strong when confident) or no call (gray). Dots: each genome's share right. Thin gray bars: the share a call of the same size placed at random would get right. Beneath: the inputs of each column, and the genomes and chromosomes it pools. In fungi and *Plasmodium* a call is right within 250 kb without Hi-C and within one Hi-C bin (10-50 kb) with it.

C Every genome with every set of inputs it was run with; each cell splits its chromosomes by outcome, and a frame marks the genome's run with the most inputs, the one D pools. *C. neoformans* appears twice, one assembly with two Hi-C libraries; B, D and E pool only the one with more read pairs (anaphase Hi-C). 5 mammals were compared with another tool's satellite locus, not with a mapped centromere, and are left out of B. The 4 holocentric genomes have no single centromere, so every call counts as wrong; of their 184 chromosomes over all sets of inputs, 0 got a confident call.

D Over the 63 genomes with a mapped centromere, each run with its most inputs, 99% of high, 88% of medium and 76% of low calls were right (calls per bar at its foot; dots: genomes); 9 chromosomes got no call.

E Calls that adding Hi-C changed (moved, or their confidence), against the same genome without Hi-C: 36% of 1,153 with real Hi-C (89 runs), 0% of 854 with Hi-C whose contacts were randomized (55 runs). Dots: genomes. Hi-C without centromere signal changes nothing. Whether real Hi-C helped: B.

Data: benchmark/results.tsv (cenfuse 0.16.0, 0.16.1) · Script: benchmark/performance.py · 2026-11-14